

Submission Summary

Conference Name

2nd International Conference on Electronics, Computing, Communication and Networking
Automation Technologies 2026

Paper ID

2446

Paper Title

Heart Disease Risk Assessment using Agentic AI

Abstract

Heart diseases are among the leading causes of death worldwide, making early detection and prevention extremely important. This paper presents a heart disease prediction system that combines machine learning and agentic artificial intelligence within a web-based healthcare application. The system predicts the likelihood of heart disease using patient parameters such as age, gender, chest pain type, blood pressure, cholesterol, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, ST depression, and ST slope.

Among multiple algorithms tested, the Random Forest classifier with 300 estimators and balanced class weights was selected due to its ability to handle mixed medical data, capture non-linear relationships, and reduce overfitting. The backend is developed using Flask REST API, which manages prediction, chat, history, and PDF report generation. The frontend is built using HTML, CSS, and JavaScript with a dark glassmorphism design and real-time 3D particle effects.

A major feature of the project is its multi-agent AI architecture powered by Groq Cloud API and large language models. Different AI agents convert patient descriptions into structured data, explain predictions in simple language, suggest lifestyle improvements, conduct follow-up interactions, maintain prediction history, and generate downloadable PDF reports.

The Random Forest model achieved 89.67% accuracy, 88.79% precision, 93.14% recall, 90.91% F1-score, and 93.50% ROC-AUC. The final system provides users with risk assessment scores, medical explanations, personalized health recommendations, AI-based chat support, and downloadable reports. This project demonstrates how machine learning and agentic AI can be integrated to create an effective healthcare risk assessment and clinical decision support system.

Keywords: Heart disease prediction, Random Forest, machine learning, agentic AI, Flask REST API, healthcare analytics, clinical decision support.

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Submission Files

Heart_Disease_RP_Format_Final (1441).pdf (548.8 Kb, 2026-05-19 12:18:48 IST)
